Welding in the World (2026) 70:4063–4077
https://doi.org/10.1007/s40194-025-02136-w

RESEARCH PAPER

Toward a validation of coupled arc-melt pool model for a TIG spot weld configuration

Marc Le Gal La Salle[1,2] · Stephen Cadiou[1] · Mickael Courtois[1] · William Berckmans[1] · Muriel Carin[1] · Alexandre Brosse[2]

Received: 24 April 2025 / Accepted: 22 July 2025 / Published online: 18 August 2025
© The Author(s) 2025

Abstract
The TIG process is a complex arc welding method due to the multitude of operating parameters and their interdependence, making it challenging to master. To improve understanding of it, a magneto-thermo-hydrodynamic model with arc-melt pool coupling is realized. A comparison is made between numerical and exhaustive experimental results of welding performed on 304L stainless steel in spot configuration at 150 A and 250 A. The effect on the melt pool of increasing the intensity is numerically studied and compared to experimental trends. A significant reduction of the model error is observed when considering the deformation of the anode-plasma interface, which strongly impacts the arc and the melt pool at high intensity. However, the numerical melt pool size is still underestimated compared to the melt areas observed on the post-mortem micrographs. The effect of an additional potential drop term that can be found in the literature is studied and demonstrated to be ineffective. The experimental melt area overestimation is discussed to explain the differences observed.

Keywords GTAW · TIG · Spot welding · Magneto-thermo-hydrodynamic · Arc plasma · Melt pool deformation · 304L stainless steel

1 Introduction

The tungsten inert gas (TIG or GTAW) process is an arc welding method commonly used in industry. However, mastering it is made complex by the multitude and interdependence of operating parameters. Simulation is one way to improve understanding and control of this process.

Multiple models exist, historically incorporating only the melt pool, such as the works of Lawson [1], Oreper [2], Szekely [3], Zacharia [4], and Debroy [5], using the finite difference method. Subsequently, works were initiated to integrate the arc with founding models developed by Hsu [6] and McKelliget [7]. Tanaka [8] was one of the first to model the whole process by integrating arc-melt pool coupling. Further advancements to this model include the introduction of metallic vapors by Zhao [9] and Lago [10], as well as the development of anode-plasma interface deformation by Lin [11], Traidia [12], and Nahed [13]. These models are influenced by multiple assumptions and uncertainties, which is why it is necessary to quantify their predictive accuracy by comparing them with experimental data. This approach has been carried out, for example, based on the work of Nestor [14] and Dupont [15], who experimentally determined the heat input generated by the arc in the welded part according to the operating parameters. Temperature measurements were also performed by Hsu [6] using spectrometry in the plasma. Tanaka [8] used these results to improve the predictive accuracy of their model by modifying the coefficients of Richardson's law. At the anode scale, the measurements made were mainly micrographs, allowing visualization of the melt pool after welding, and temporal temperature measurements. Multiple authors such as Tanaka [8], Brochard [16], and Traidia [12] present these results for stainless steels with different chemical compositions and for various process parameters. However, for these numerical studies, comparisons with experimental data remain limited, often focusing

Recommended for publication by Commission II - Arc Welding and Filler Metals

✉ Marc Le Gal La Salle
marc.le-gal-la-salle@framatome.com

[1] Université Bretagne-Sud, UMR CNRS 6027, IRDL, F-56100 Lorient, France

[2] Framatome-DTIM, 2 Rue Professeur Jean Bernard, 69007 Lyon, France

on cases with low intensities, leading to minimal deformations of the free surface of the melt pool.

This paper presents experimental results on melt pools and temperature measurements at 150 and 250 A on the same batch of 304L stainless steel, ensuring identical thermophysical properties, including the thermocapillary coefficient, which is responsible for the Marangoni effect, across all experiments. These experimental results are compared with those obtained through simulation performed with COMSOL Multiphysics®. Both numerical and experimental observations show an enlargement of the melt pool with increased intensity. The width-depth ratio doubles experimentally and increases by a factor of 1.7 in simulation, resulting in a relative error of 6.5%, demonstrating the simulation's ability to predict the influence of intensity on the melt pool behavior. This evolution is attributed to the significant deformation of the anode-plasma interface, whose effect on the plasma and melt pool is newly exposed. Additionally, an underestimation of the melt pool volume is observed, which justifies the improvement of the model by two distinct methodologies. The first involves adding additional heat input by considering another potential drop in the anodic layer, as proposed by Kaddani [17]. The second approach, more physical, but degrading numerical stability, is to decrease the viscosity of the liquid metal to tend toward a realistic value. The results of these two solutions are presented and compared to determine the perspectives of this model.

2 Welding configuration studied

A melt pool is produced using the TIG process at the center of a stainless-steel cylinder (304L), as shown in Fig. 1. The welded part is supported at its periphery by three pins to minimize the effect of the support on the heat transfer of the sample. These pins are connected from the positive pole of the welding machine to ensure relative axisymmetric of the electric field circulation within the arc and the melt pool. The arc is formed at a constant height of 2 mm between the upper face of the cylinder and a thoriated tungsten electrode (WT20). The absence of vertical movement of the torch necessitates high-frequency starting. The electrode is meticulously sharpened for each weld to maximize the axiality of its scratches and to reduce dispersion associated with its geometric characteristics. The stick-out is intentionally significant to improve visibility of the melt pool. The shielding gas consists of pure industrial argon (air liquid argon $\geq$ 99.99%, $H_2O \leq 40$ ppmv) injected at a flow rate of 15 L/min through a nozzle without a sieve. The geometrical parameters associated with this configuration are shown in Fig. 2. For this work, intensities of 150 and 250 A are studied. The current profile is detailed in Fig. 3, starting at 100 A and ramping up over 0.5 s to reach the target value of 150 or 250 A, which is maintained for 9.5 s before the current is abruptly stopped.

3 Experimental results

The previously described experimental setup is used to perform four welds at 150 and 250 A to acquire experimental data, including micrographs of the melt pool, in-situ

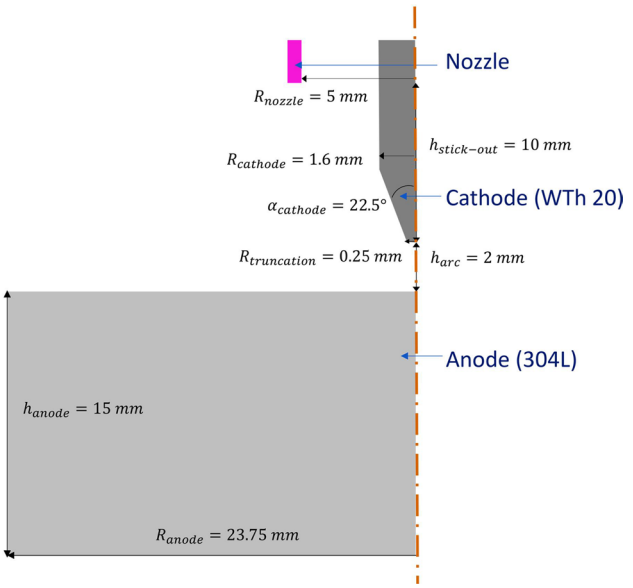


Fig. 2 Schematic representation of the device

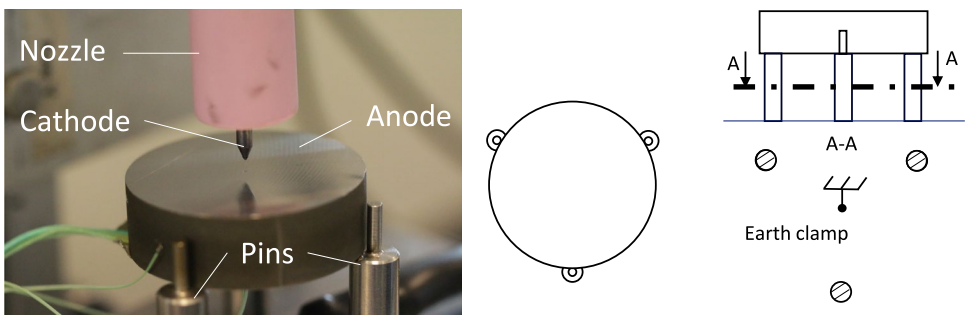


Fig. 1 Photography and technical drawing of the experimental setup

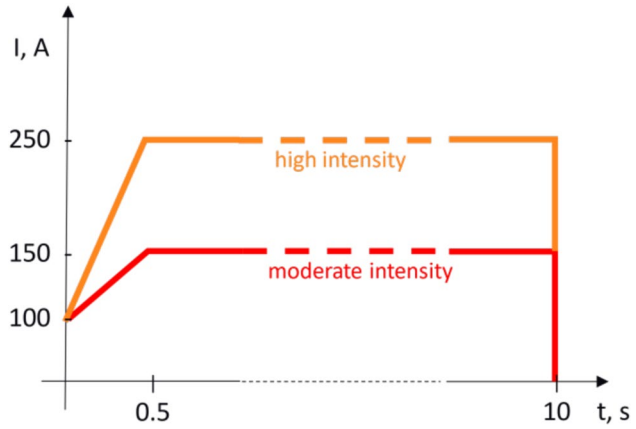


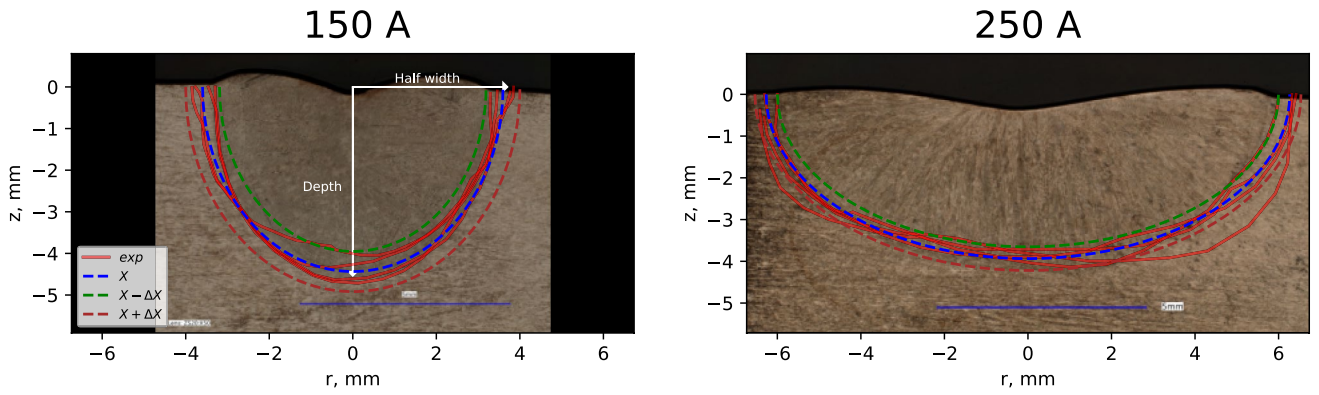
Fig. 3 Temporal evolution of applied current

temperature measurements, images of the melt pool surface during welding, and voltage measurements. The objective of this section is to present and analyze these data.

Micrographs are obtained by cutting through the center of the weld, polishing, and etching with Marble's reagent to highlight the outline of the melt pool at $t = 10$ s. These micrographs are used to determine the geometric characteristics of the melt pool as summarized in Fig. 4. A statistical analysis of these results is performed to deduce uncertainty using the A-type method for each intensity. It is observed that increasing the intensity affects the melt pool in several ways. First, its volume increases, which is an expected result. Second, its shape is modified, resulting in an approximate doubling of the half-width to depth ratio. Third, the experimental dispersion is slightly lower at high intensity reflecting a stabilization of the phenomena acting on the melt pool.

The temporal evolution of the temperature within the anode is measured using K-type micro-thermocouples during welding at 150 A. These thermocouples (wires of 50 µm) are placed as close to the melt pool as possible by micro-drilling holes with a diameter of 0.65 mm. The associated micrograph and the measured temperatures are shown in Fig. 5. The start of welding causes a rapid rise in temperature, followed by a slight slowdown in this increase under the effect of diffusion within the anode. Temperature oscillations are observed above 1000 K between 5 and 10 s of welding. The amplitudes of the oscillations increase with temperature, reflecting a local instability of temperature and therefore potentially of the position of the fusion isotherm. The amplitudes are proportionally lower for TC2 because it is further from the melt pool. There is also a similar phase shift for both thermocouples which are on the same side of the weld. The arc stops at 10 s followed by a free cooling phase.

To get the thermophysical properties, a characterization campaign of the material is performed using differential scanning calorimetry (LABSYS evo STA, Setaram), thermal diffusivimeter (LFA 475/2/6, Netzsch), and a dilatometer (L75/20C, Linseis). These results are used to adjust the properties exposed by Kim [18], as shown in Fig. 6 and Eqs. (1) to (9) for the steel used in this study. In addition, the solidus is measured at 1708 K and the liquidus at 1740 K.



	Half width, mm	Depth, mm	Volume, mm³
1	3.77	4.05	118.2
2	3.46	4.34	107.6
3	3.31	4.64	117.6
4	3.85	4.71	124.8
X	3.6±0.4	4.44±0.48	117.1±11.3

	Half width, mm	Depth, mm	Volume, mm³
1	6.02	4.14	315.0
2	6.32	4.0	388.0
3	6.42	3.9	359.9
4	6.31	3.71	329.3
X	6.27±0.27	3.94±0.28	348.1±51.8

Fig. 4 Micrographs at 150 and 250 A

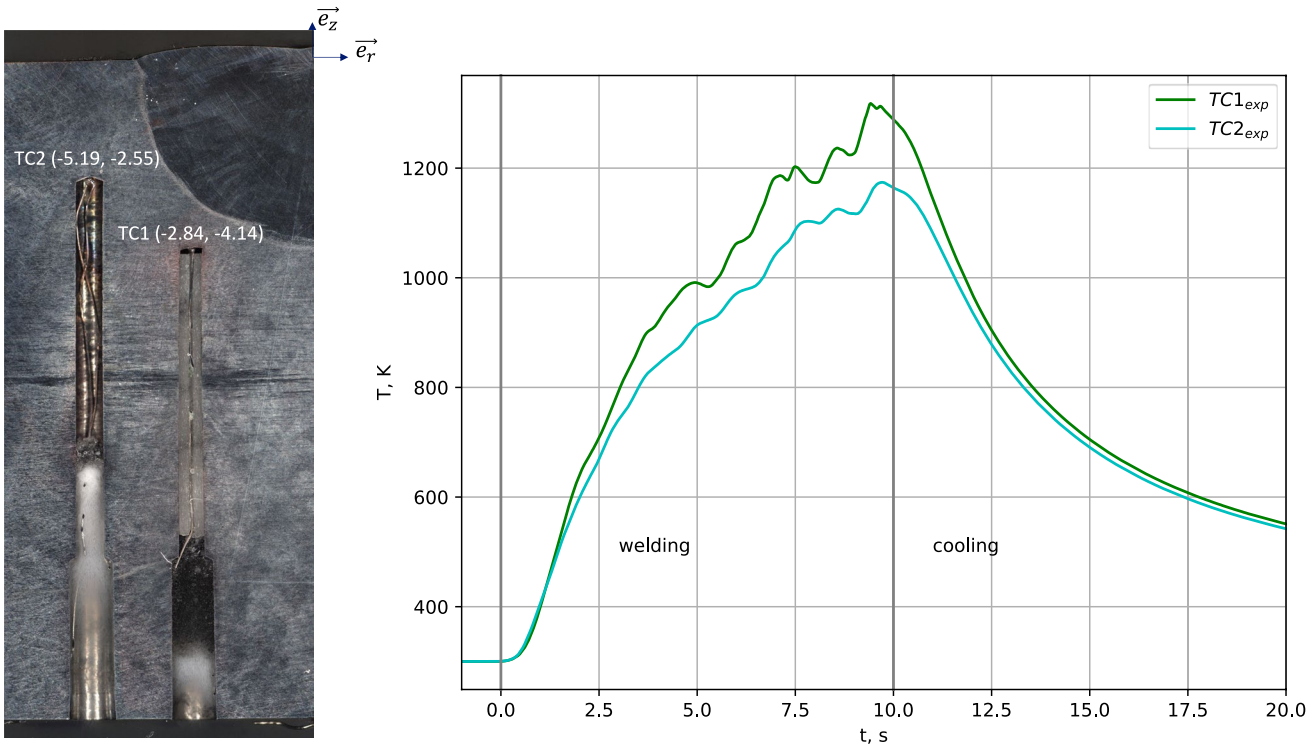


Fig. 5 Micrography of a weld with micro-thermocouples and measurements associated

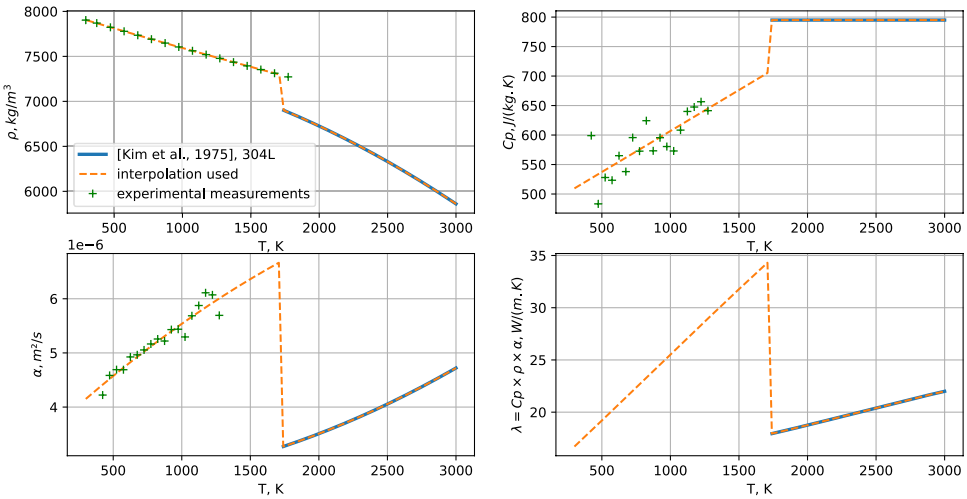


Fig. 6 Thermophysical properties of 304L stainless steel

$$\rho_s = 8.04.10^3 - 4.62.10^{-1} \times T + 1.69.10^{-5} \times T^2 \quad (1)$$

$$\rho_l = 7.55.10^3 - 1.12.10^{-1} \times T - 1.51.10^{-4} \times T^2 \quad (2)$$

$$\rho = \rho_s \times (1 - fl) + \rho_l \times fl \quad (3)$$

$$Cp_s = 4.68.10^2 + 1.39.10^{-1} \times T \quad (4)$$

$$Cp_l = 7.95.10^2 \quad (5)$$

$$Cp = Cp_s \times (1 - fl) + Cp_l \times fl \quad (6)$$

$$\lambda_s = 13.1 + 1.19.10^{-2} \times T + 8.75.10^{-7} \times T^2 - 3.72.10^{-10} \times T^3 + 2.11.10^{-14} \times T^4 - 6.49.10^{-19} \times T^5 \quad (7)$$

$$\lambda_l = 1.51 - 1.03.10^{-4} \times T + 1.13.10^{-6} \times T^2 - 2.36.10^{-11} \times T^3 - 2.86.10^{-14} \times T^4 \quad (8)$$

$$\lambda = \lambda_s \times (1 - fl) + \lambda_l \times fl \quad (9)$$

A video at 150A is recorded with a high-speed camera (Phantom VEO-E 310L) and an illumination laser (Cavitar Cavilux Smart) to enhance the visibility of the surface of the melt pool. Figure 7 is a frame of this video that allows a qualitative observation of the surface flows and the deformation of the anode-plasma interface.

Voltage is measured by the welding station (Fronius Magicwave 3000) and shown in Table 1. This potential difference includes cables and connections, which may cause a slight overestimation of the one between the top of the cathode and the bottom of the anode.

Table 1 Voltage measurements

I, A	U, V(t = 2.5 s)
150	11.2
250	12.8

4 Magneto-thermo-hydrodynamic simulation of the TIG process

4.1 Phenomena involved

A magneto-thermo-hydrodynamic model with surface deformation in ALE and melt pool-arc coupling in a 2D axisymmetric configuration has been developed to simulate the previous experimental setup. Figure 8 summarizes all the physical phenomena in this simulation.

4.2 Mathematical formulation

This formulation is inspired and validated by the numerical and experimental works of Hsu [6], Tanaka [8], Brochard [16], Traidia [12], Cadiou [19], and Nahed [13] (Table 2).

The nomenclature and equations associated with this model are detailed in Tables 2 and 3 for electromagnetism, fluid flow in plasma and melt pool, heat transfer, and anode-plasma interface deformation. For simplicity, the axisymmetric conditions are not included. These equations apply to the geometric domains and boundaries as shown in Fig. 9. Figure 10 summarizes the couplings between these different equations. The material properties used are detailed in Tables 4 and 5. Multiple assumptions are made in addition to axisymmetric, on the scale of the arc:

- local thermodynamic equilibrium (LTE): homogenization of ion and electron temperatures within the arc column, excluding the cathodic and anodic interfaces by using modified electrical conductivity as described by Lowke [20];
- quasi-neutrality;
- Thomson's effect is regarded as negligible;
- simplified modeling of the anodic interface with an associated source term, as described by Tanaka [8];
- approximation of radiative losses by a net emission coefficient assuming plasma transparent and optically thin without interaction with electrodes surfaces;
- hydrodynamic continuity, laminar flow, and weakly compressible plasma;
- plasma composed of pure argon without metallic vapors.

At the scale of the melt pool, the liquid metal is considered an incompressible laminar flow. To obtain a zero-velocity field in the solid phase, an artificially high viscosity is chosen for temperatures below the melting point, along with the addition of a Darcy term in the momentum equations. To

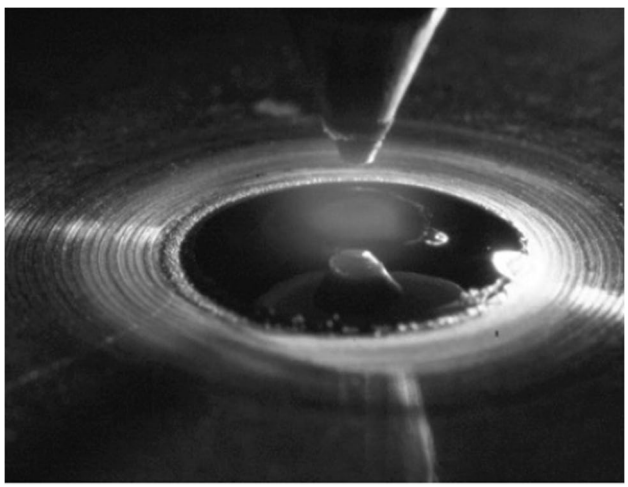


Fig. 7 High-speed camera frame of the melt pool at moderate intensity

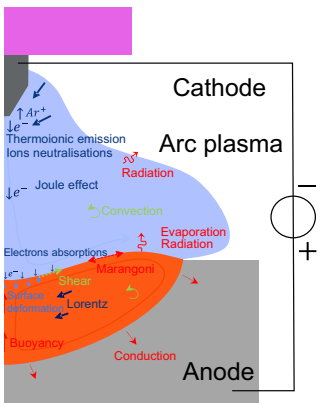


Fig. 8 Physical phenomena considered in simulation

Table 2 Nomenclature

$\vec{A}$	Magnetic vector potential	$\vec{n}$	Normal vector
A_r	Richardson constant	∇_s	Surface divergence
B	Darcy's law coefficient	p_1	Pressure in plasma
$\vec{B}$	Magnetic flux density	p_2	Pressure in anode
C	Darcy's law coefficient	ϕ	Axial displacement of the mesh
C_p	Heat capacity at constant temperature	ϕ_a	Work function at the anode interface
e	Elementary charge	ϕ_e	Work function associated with thermoionic emission
$\vec{E}$	Electric field	ϕ_c	Work function at the cathode interface
ϵ	Net emissivity	ψ	Axial displacement of the interface anode-plasma
ϵ_a	Anode surface emissivity	$\vec{q}$	Heat flux vector
ϵ_c	Cathode surface emissivity	Q_{anode}	Total heat flux at anode surface
fl	Liquid fraction of metal	σ	Electrical conductivity
$\vec{g}$	Acceleration of gravity	σ_B	Stefan-Boltzmann constant
γ	Surface tension coefficient for Marangoni	$\overline{\overline{\sigma_1}}$	Stress viscous tensor of plasma
γ^*	Equivalent surface tension coefficient for interface deformation	$\overline{\overline{\sigma_2}}$	Stress viscous tensor of anode
γ_l	Equivalent surface tension coefficient for interface deformation	ρ	Density
γ_s	Equivalent surface tension coefficient at solid state for metal	ρ_{ref}	Reference density for thermal expansion of liquid metal
$\vec{H}$	Magnetic density	s_{vap}	Vaporization source term
h_{conv}	Convection coefficient	T	Temperature
I_0	Input intensity	$\vec{t}$	Tangential vector
$\vec{J}$	Current density vector	T_{ref}	Reference temperature for thermal expansion of liquid metal
j_c	Normal current density at cathode surface	T_s	Metal solidus
j_e	Current density associated with thermionic emission	T_l	Metal liquidus
j_{em}	Current density associated with Richardson's law	T_0	Ambient temperature
j_{ion}	Current density with ion neutralization	U_{plasma}	Tension without electrodes interfaces
k_B	Boltzmann constant	U_{soudage}	Total tension
L_{vap}	Iron vaporization enthalpy	$\overrightarrow{u_1}$	Velocity in plasma
λ	Thermal conductivity	$\overrightarrow{u_2}$	Velocity in anode
μ	Dynamic viscosity	V	Electrical potential
μ_0	Absolute permeability	V_i	Argon ionization potential
μ_r	Relative permeability		

facilitate the convergence of the calculations, the dynamic viscosity of the liquid phase is enhanced and set to 0.03 Pa.s, values consistent with those of Brochard [16], Traidia [12], and Nguyen [21], although more realistic values range from 0.0015 to 0.006 Pa.s [22]. The external heat dissipation of the anode is modeled by convection and radiation whose coefficients have not been adjusted as they have very small influence on the heating phase and therefore the melt pool. The fusion enthalpy is not taken into account based on the work of Cadiou [19] in spot configuration.

4.3 Numerical resolution

The time scheme employed to solve the previously mentioned equations is the second-order backward differentiation formula (BDF) with Anderson acceleration. This method is applied in a segregated way to separate the physics of the plasma, the melt pool, and the interface deformation. The error criterion used in COMSOL Multiphysics® is defined by Eq. (59), which evaluates iterative stability with M as the number of fields, N_j as the number of degrees of freedom for field j, W_j as the scale, and $U_{i,j}$ as the solution vector. Calculations are repeated for all variables until the relative error falls below 0.1%. The mesh used consists of linear tetrahedral elements, whose arrangement and dimensions are shown in Fig. 11. Its size is chosen to finely describe the flows while maintaining a laminar regime.

$$err = \sqrt{\frac{1}{M}\sqrt{\sum_{j=1}^{M}\frac{1}{N_J}\sum_{i=1}^{N_j}\left(\frac{\left|U_{i,jt}-U_{i,jt}-\Delta t\right|}{max\left(\left|U_{i,jt}\right|\right)}\right)^2}} \tag{59}$$

Table 3 Model equations

Electromagnetism $\left(\Omega_c \cup \Omega_p \cup \Omega_{al} \cup \Omega_{as}\right)$		
$\vec{B} = \nabla \times \vec{A}$	(10)	$\Omega \in (\Omega_c \cup \Omega_p \cup \Omega_{al} \cup \Omega_{as})$
$\vec{B} = \mu_r \mu_0 \vec{H}$	(11)	
$\vec{E} = -\nabla V$	(12)	
$\nabla \times \vec{H} = \vec{J}$	(13)	
$\vec{J} = \sigma \vec{E}$	(14)	
$\nabla . \vec{J} = 0$	(15)	
$\nabla . \vec{A} = 0$	(16)	
$\int -\vec{J} . \vec{e_z} dS = I_0$	(17)	$\partial\Omega \in AB$
$V = 0$	(18)	$\partial\Omega \in GF$
$\vec{n} . \vec{J} = 0$	(19)	$\partial\Omega \in (BD \cup DF \cup FG)$
$\vec{n} \times \vec{A} = \vec{0}$	(20)	
$j_c = \vec{J} . \overrightarrow{n_{\text{cathode}\rightarrow\text{plasma}}}$	(21)	$\partial\Omega \in (\partial\Omega_c \cap \partial\Omega_p)$
$j_{\text{em}} = A_r T^2 \exp\left(-\frac{e\phi_e}{k_B T}\right)$	(22)	
$j_{\text{ion}} = \begin{cases} j_c - j_{\text{em}} for j_c - j_{\text{em}} > 0 \\ 0 otherwise \end{cases}$	(23)	
$j_e = j_c - j_{\text{ion}}$	(24)	
Heat transfer $\left(\Omega_{\text{c}} \cup \Omega_{\text{p}} \cup \Omega_{\text{al}} \cup \Omega_{\text{as}}\right)$		
$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \vec{u} . \nabla T + \nabla . \vec{q} = S$	(25)	$\Omega \in (\Omega_c \cup \Omega_p \cup \Omega_{al} \cup \Omega_{as})$
$\vec{q} = -\lambda \times \nabla T$	(26)	
$S = \begin{cases} \vec{J} . \vec{E} - 4\pi\epsilon for \Omega \in \Omega_p \\ \vec{J} . \vec{E} for \Omega \in \Omega_{al} \cup \Omega_{as} \cup \Omega_c \end{cases}$	(27)	
$T = T_0$	(28)	$\partial\Omega \in AB$
$\vec{q} . \vec{n} = 0$	(29)	$\partial\Omega \in (BD \cup DE)$
$-\vec{n} . \vec{q} = h_{\text{conv}} \times \left(T_0 - T\right) + \epsilon_a \sigma_B \times (T_0^4 - T^4)$	(30)	$\partial\Omega \in (EF \cup FG)$
$-\vec{n} . \vec{q} = j_{ion} V_i - j_e \phi_c - \epsilon_c \sigma_B T^4$	(31)	$\partial\Omega \in (\Omega_p \cap \Omega_c)$
$-\vec{n} . \vec{q} = \vec{J} . \vec{n} \phi_a - \epsilon_a \sigma_B T^4 - s_{vap}$	(32)	$\partial\Omega \in (\partial\Omega_p \cap (\partial\Omega_{al} \cup \partial\Omega_{as}))$
$S_{vap} = -10\left(2.52 + \left(6.121 - \frac{18836}{T}\right) - 0.5\log_{10}(T) + 1\right) \times L_{vap}$ [25]	(33)	
Fluid flow in plasma (Ω_{p})		
$\frac{\partial \rho}{\partial t} + \nabla . \left(\rho \overrightarrow{u_1}\right) = 0$	(34)	$\Omega \in \Omega_p$
$\rho\left(\frac{\partial \overrightarrow{u_1}}{\partial t} + \left(\overrightarrow{u_1} . \nabla\right)\overrightarrow{u_1}\right) = \nabla . \left(-p_1 \bar{\bar{I}} + \bar{\bar{\sigma}}\right) + \vec{J} \times \vec{B} + \rho \vec{g}$	(35)	
$\bar{\bar{\sigma}} = \mu\left(\nabla \overrightarrow{u_1} + \nabla \overrightarrow{u_1}^T\right) - \frac{2}{3}\mu\left(\nabla . \overrightarrow{u_1}\right)\bar{\bar{I}}$	(36)	
$p_1 = 0$	(37)	$\partial\Omega \in D$
$\overrightarrow{u_1} = \vec{0}$	(38)	$\partial\Omega \in (\partial\Omega_p \cap (\partial\Omega_{al} \cup \partial\Omega_{as} \cup \partial\Omega_c)$
$\bar{\bar{\sigma}} . \vec{n} = \vec{0}$	(39)	$\partial\Omega \in (CD \cup DE)$
$\overrightarrow{u_1} = -\frac{2Q_v}{\pi} \times \frac{R_{\text{nozzle}}^2 - r^2 + \left(R_{\text{nozzle}}^2 - R_{\text{cathode}}^2\right) \times \log\left(\frac{r}{R_{\text{nozzle}}} - \frac{R_{\text{nozzle}}}{R_{\text{cathode}}}\right)}{R_{\text{nozzle}}^4 - R_{\text{cathode}}^4 + \frac{\left(R_{\text{nozzle}}^2 - R_{\text{cathode}}^2\right)^2}{\log\left(\frac{R_{\text{nozzle}}}{R_{\text{cathode}}}\right)}} \overrightarrow{e_z}$	(40)	$\partial\Omega \in BC$
$\overrightarrow{u_1} . \vec{t} = 0$	(41)	$\partial\Omega \in (\partial\Omega_p \cap \partial\Omega_a)$
$\overrightarrow{u_1} . \overrightarrow{e_z} = \frac{\partial \psi}{\partial t}$	(42)	

Table 3 (continued)

Fluid flow in melt pool (Ω_{al})		
$\rho\nabla.\overrightarrow{u_2} = 0$	(43)	$\Omega \in \Omega_{al}$
$\rho\left(\frac{\partial\overrightarrow{u_2}}{\partial t} + \left(\overrightarrow{u_2}.\nabla\right)\overrightarrow{u_2}\right) = \nabla.\left(-p_2\bar{\bar{I}} + \bar{\bar{\sigma}}\right) + \vec{F}$	(44)	
$\bar{\bar{\sigma}} = \mu\left(\nabla\overrightarrow{u_2} + \nabla\overrightarrow{u_2}^T\right)$	(45)	
$\vec{F} = \vec{J} \times \vec{B} - \rho_{\text{ref}}\beta\left(T - T_{\text{ref}}\right)\vec{g} - C\frac{(1-fl)^2}{fl^3+B}\overrightarrow{u_2}$	(46)	
$fl = \begin{cases} 0 for t < T_s \\ \frac{T-T_s}{T_l-T_s} for T_s \le T \le T_l \\ 1 for T > T_l \end{cases}$	(47)	
$p_2 = 0$	(48)	$\partial\Omega \in I$
$\bar{\bar{\sigma}}.\vec{t} = \frac{\partial\gamma}{\partial T}(\nabla T - \nabla T.\vec{n}) + \mu_1\nabla(\overrightarrow{u_1}.\vec{t}).\vec{n}$	(49)	$\partial\Omega \in \left(\partial\Omega_p \cap \partial\Omega_{al}\right)$
$\overrightarrow{u_2}.\vec{n} = \frac{\partial\psi}{\partial t}$	(50)	
$\overrightarrow{u_2} = \vec{0}$	(51)	$\partial\Omega \in \left(\partial\Omega_{al} \cap \partial\Omega_{as}\right)$
Movement of the anode-plasma interface ($\partial\Omega_{al} \cap \partial\Omega_p$)		
$\begin{pmatrix} \left(1-n_r^2\right)\frac{\partial}{\partial r} - n_r n_z \frac{\partial}{\partial z} \\ -n_r n_z \frac{\partial}{\partial r} + \left(1-n_z^2\right)\frac{\partial}{\partial z} \end{pmatrix}.\left(\frac{\gamma^*}{\sqrt{1+\frac{\partial\psi^2}{\partial r^2}_T}}\right) = p_1$	(52)	$\partial\Omega \in (\partial\Omega_{al} \cap \partial\Omega_p)$
$\gamma^* = \gamma_l \times fl + \gamma_s \times (1 - fl)$	(53)	
$\int \psi = 0$	(54)	
$\psi = 0$	(55)	$\partial\Omega \in J$
Moving mesh ($\Omega \in \Omega_{\text{ALE}}$)		
$\frac{\partial}{\partial z}\left(\frac{1}{z}\frac{\partial\phi}{\partial z}\right) = 0$	(56)	$\Omega \in \Omega_{\text{ALE}}$
$\phi = 0$	(57)	$\partial\Omega \in (PQ \cup QO \cup ON)$
$\phi = \psi$	(58)	$\partial\Omega \in (\partial\Omega_{al} \cap \partial\Omega_p)$

5 Numerical analysis of the effect of increasing intensity

Experimental observations indicate that increasing the intensity significantly alters the melt pool's characteristics. Firstly, the volume increases due to the rise in current and voltage for a constant arc length. In simulation, the potential difference between the top of the cathode and the bottom of the anode changes from 10.6 to 11.5 V. This numerical evolution qualitatively matches the experimental trend, despite a relative error of 5 and 10.2%. Besides a natural increase in volume, its morphology is notably affected. The half-width-to-depth ratio changes from 0.81 at 150 A to 1.59 at 250 A, indicating an enlargement of the melt pool. The objective of this part is to compare this experimental tendency and that associated with the model previously presented.

The temperature derivative of the surface tension coefficient ($\frac{\partial\gamma}{\partial T}$) has a significant influence on the flows in the melt pool and, consequently, its shape due to the Marangoni effect. Determining this value is complex because it is affected by multiple chemical species such as sulfur, oxygen, manganese, etc. In the absence of experimental characterization, this value is extrapolated in simulation to fit with the experimental shape of melt pool at 150 A. An optimal result is obtained for $\frac{\partial\gamma}{\partial T} = 1.7.10^{-4}$ N/(m.K).

The melt pools at 150 and 250 A after 10 s of welding obtained with simulation are shown in Fig. 12. This enlargement of the melt pool is due to an increase in heat input and more specifically to the action of the arc pressure at the anode-plasma interface, resulting in greater deformation and causing a radial ejection of fluid. Moreover, mass conservation was properly respected for both case studies at 150 and 250 A. This primarily explains the numerically observed enlargement of the melt pool. This trend is similar to experimental observations, where the half-width-to-depth ratio doubles with an intensity change from 150 to 250 A, while in simulation it is 1.87 with surface deformation. These values are very close, confirming the simulation's ability to predict the influence of this process parameter.

The satisfactory predictive accuracy at high intensity is due to the implementation of surface deformation in the simulation. This paragraph aims to explain the differences generated by considering surface deformation at the scale of the melt pool and the arc. Figure 13 compares the simulation results with and without surface deformation for both intensities. The differences are almost negligible at 150 A

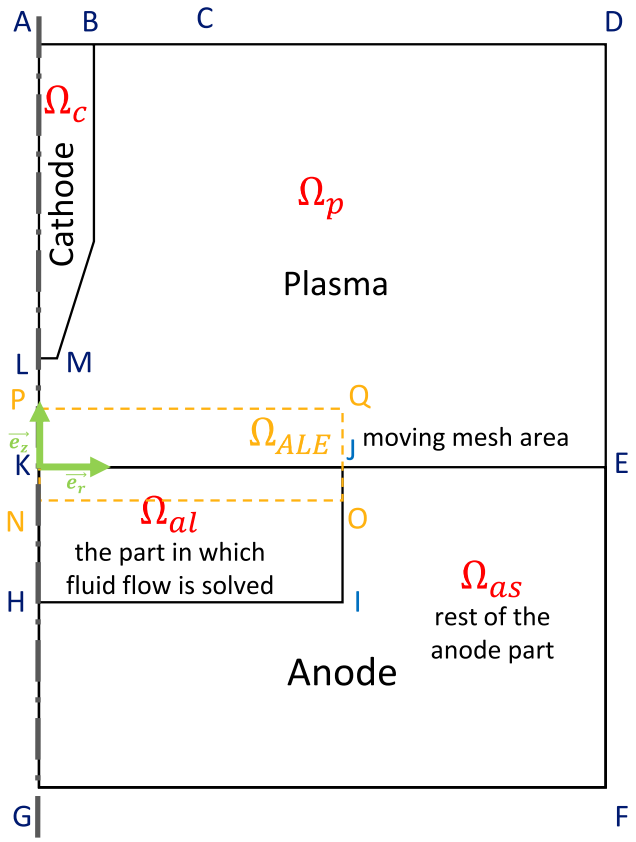


Fig. 9 Solving domains

but become significant at 250 A, where the relative difference in the half-width-to-depth ratio is 25%. At high intensity, the numerical result with deformation is much closer to the experimental result, with a relative difference of 6.5%

Table 4 Material properties

	Cathode (Ω_c)	Plasma (Ω_p)	Anode ($\Omega_{al} \cup \Omega_{as}$)
ρ	$17,800 \mathrm{kg/m^3}$	[23]	Figure 4 and Equation (3)
C_p	$132 \mathrm{J/(kg.K)}$		Figure 4 and Equation (6)
λ	[24]		Figure 4 and Equation (9)
ϵ	∅		∅
μ	∅		$\begin{cases} 10^4 Pa.s \text{ for } T < T_f \\ 3.10^{-2} Pa.s \text{ for } T > T_f \end{cases}$
σ	[18]	$\sigma_{tungsten}$ [18] near cathode σ_{argon} [23] $7.7.10^5$ S/m near anode	$7.7.10^5 \mathrm{S/m}$
μ_r	1	1	1

compared to 28% without deformation. These figures justify the need to consider surface deformation at 250 A; however, its consideration is not essential at 150 A for this arc length.

This deformation also affects the arc. Figure 14 shows the velocity fields and temperature isovalues for an arc with and without surface deformation at 250 A. At the arc scale, surface deformation causes a radial deviation of the current lines, a decrease in the maximum axial velocity by a factor of 1.4, and a slight cooling of the plasma. These modifications directly affect the behavior of the melt pool, mainly through radial ejection caused by the sinking of its upper surface. As shown in Fig. 15, the conditions at the anode-plasma interface are significantly altered. The shear amplitude is reduced, and its distribution changes. The deviation of current lines within the arc alters the normal current density at this interface, impacting the heat input generated by electron absorption and the distribution of Lorentz forces in the melt pool.

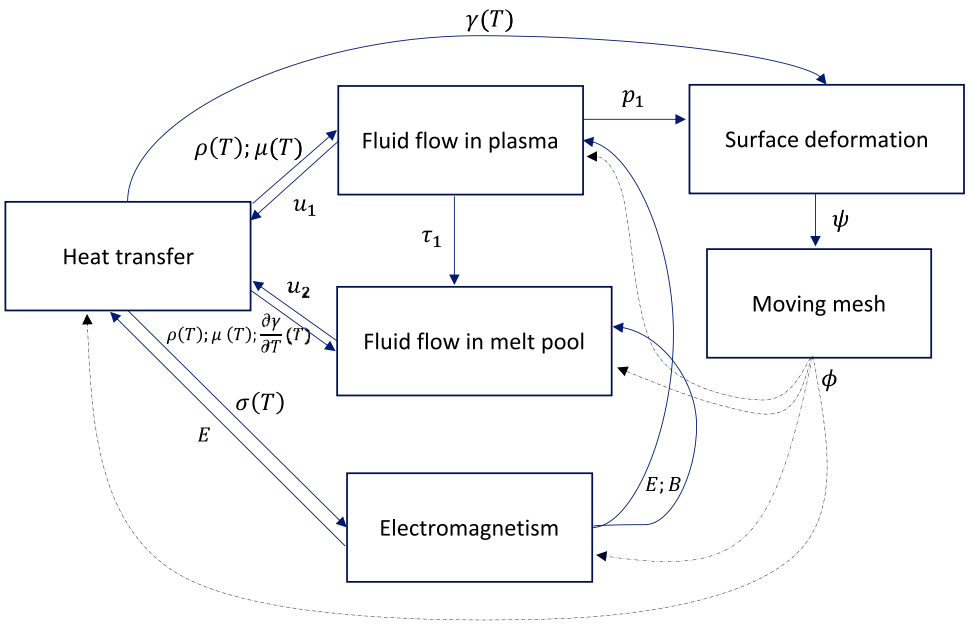


Fig. 10 Schematic representation of the couplings between equations

Table 5 Material properties

Material constant	Value
T_s	1708 K
T_l	1740 K
C/B	10^9 kg/s
ρ_{ref}	6874 kg/m^3
β	10^{-4} K^{-1}
T_{ref}	1723 K
γ	1.5 N/m [25]
h_{conv}	10 W/(m^2.K)
ϵ_c	0.4
ϵ_a	$0.7 \times fl + 0.4 \times (1 - fl)$
ϕ_e	2.63 V [6]
V_i	15.68 V [6]
ϕ_c	4.52 V [6]
ϕ_a	4.7 V [6]

In conclusion, considering surface deformation improves predictive accuracy at high intensity. The differences generated by this deformation on the arc and their impact on the melt pool justify the importance of coupling the arc and melt pool.

6 Recalibration of the numerical model

Although the model captures the evolution of the melt pool shape under increased intensity, differences in volumes and temperatures necessitate model adjustment. This recalibration phase is performed only at 150 A because there are no thermocouples at 250 A, and simulations can be conducted without considering surface deformation, reducing calculation time. Initially, the differences between numerical and experimental results at 150 A are presented. Subsequently,

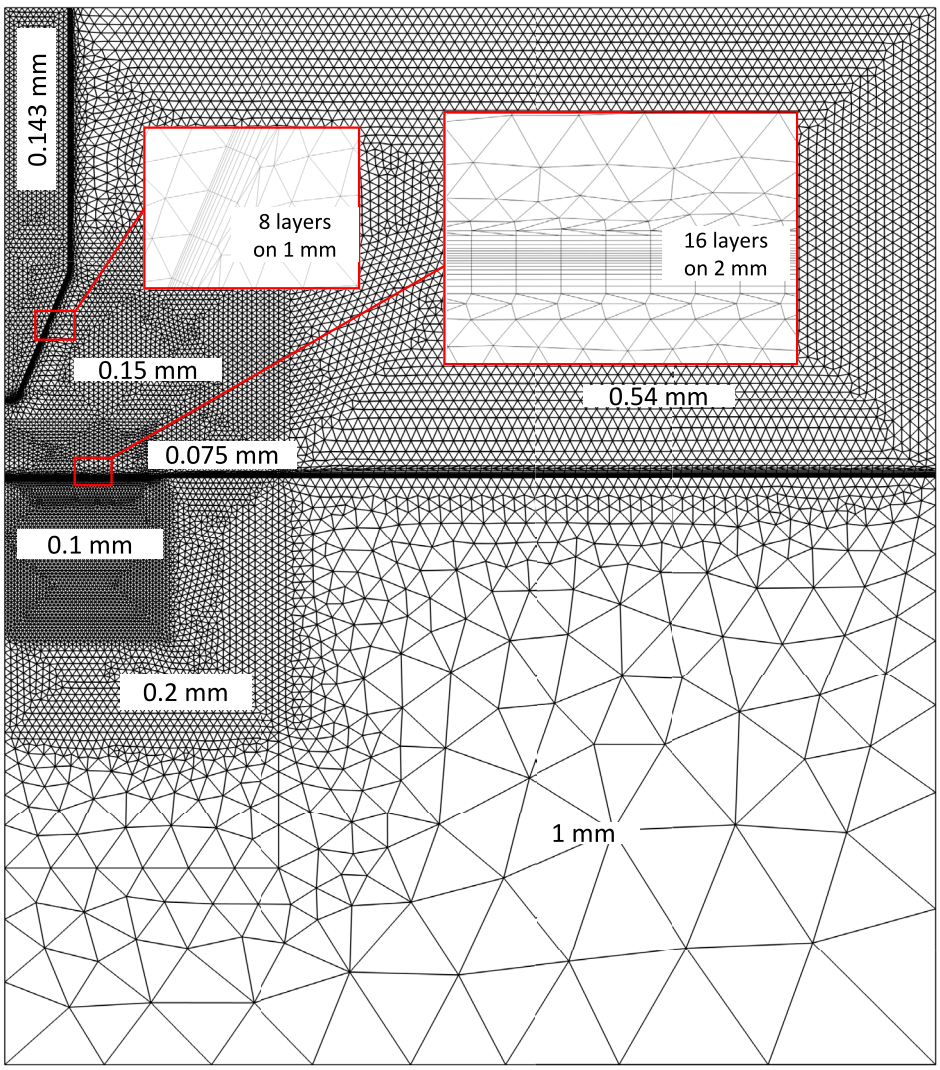


Fig. 11 Mesh sizes

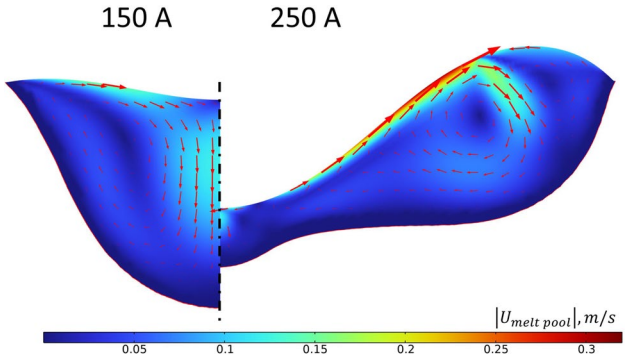


Fig. 12 Velocity into melt pool at 150 and 250 A

the results after adding a complementary potential difference in the anodic layer are discussed.

6.1 Comparison before recalibration

Figure 16 shows a numerical-experimental comparison of the melt pool and the temporal evolution of temperatures. The temperature measurements are very close below 4 s to the simulation, with a relative error of less than 2%. After that, the differences increase with time to be equal to 10% at 10 s, reflecting an underestimation of the heat source in simulation. In addition, the numerical results do not include the temperature oscillations experimentally observed. Concerning the post-mortem micrographs, there is a 40% difference between the melt volume and that of the numerical melt pool at 10 s.

6.2 Anodic potential drop recalibration

Some works in the literature introduce an additional heat input at the anode-plasma interface by an anodic potential drop (ΔV_a). This constant represents the potential heating generated by the transit of electrons in the anodic layer before their absorption. This correction addresses the inadequate description of phenomena within this layer, which arises from simplifying assumptions such as the homogenization of temperature between ions and electrons. Kaddani [17] experimentally determined ΔV_a to be 3.5 V for welding with an intensity of 200 A, an arc length of 10 mm, and a non-fusible copper anode. This value is used in simulation by Borel [26] and Brochard [16] but not by Tanaka [8], who consider it included in the work function at the anode-plasma interface ($\phi_a = 4.7$ V). The objective of this section is to estimate ΔV_a to match the experimental melt volume with the numerical melt pool at 10 s and then analyze the effect of this change.

The determination of this value is complex because it also depends on the calibration of the thermocapillary coefficient $\frac{\partial \gamma}{\partial T}$ to maintain the same shape and the calculation time of approximately 12 h. To address this issue, a model is realized without surface deformation and without arc solving. The heat flux, current density, and shear from the arc at 5 s are projected onto the anode-plasma interface to solve physics only in the anode allowing a calculation time of approximately 1 h. This model is used to determine the volume (V) and the half-width-to-depth ratio ($\frac{Hw}{D}$) of the melt pool for different values of ΔV_a and $\frac{\partial \gamma}{\partial T}$, as shown in Fig. 17. The results of 56 simulations are used to interpolate V and $\frac{Hw}{D}$ by two bivariate quadratic polynomials, as defined by Eqs. (60) and (61). The optimization to match the experimental melt pool criteria is defined by Eq. (62) and Fig. 17, and the solution found is $\Delta V_a = 5.32$ V and $\frac{\partial \gamma}{\partial T} = +1.5135.10^{-4}$ N/(m.K) at 150 A.

$$V = f_1\left(\Delta V_a, \frac{\partial \gamma}{\partial T}\right) \tag{60}$$

$$\frac{Hw}{D} = f_2\left(\Delta V_a, \frac{\partial \gamma}{\partial T}\right) \tag{61}$$

$$\begin{cases} V_{exp} = f_1\left(\Delta V_a, \frac{\partial \gamma}{\partial T}\right) \\ \frac{Hw}{D}_{exp} = f_2\left(\Delta V_a, \frac{\partial \gamma}{\partial T}\right) \end{cases} \tag{62}$$

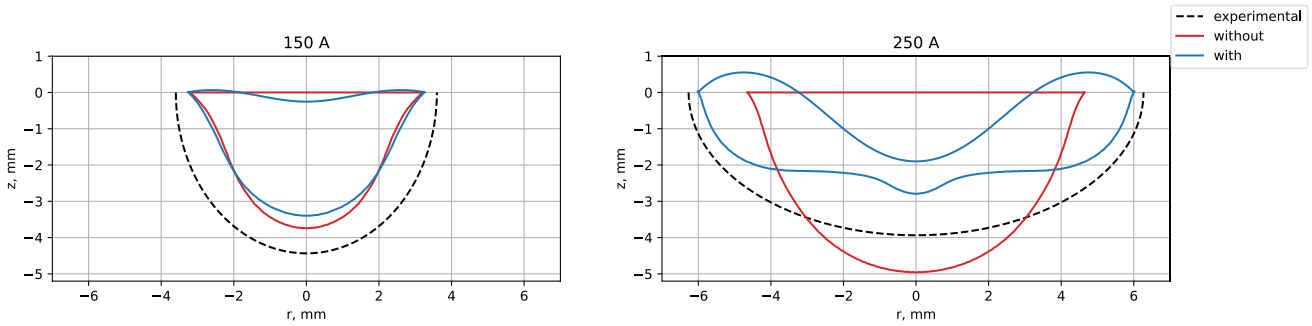


Fig. 13 Fusion isotherms with and without surface deformation for 150 and 250 A

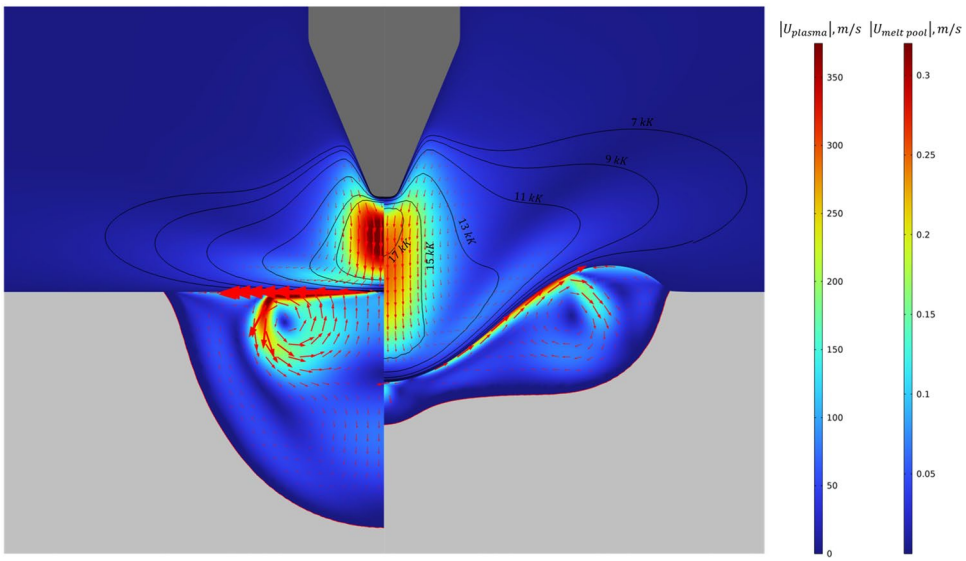


Fig. 14 Arc and melt pool without and with surface deformation at 250 A

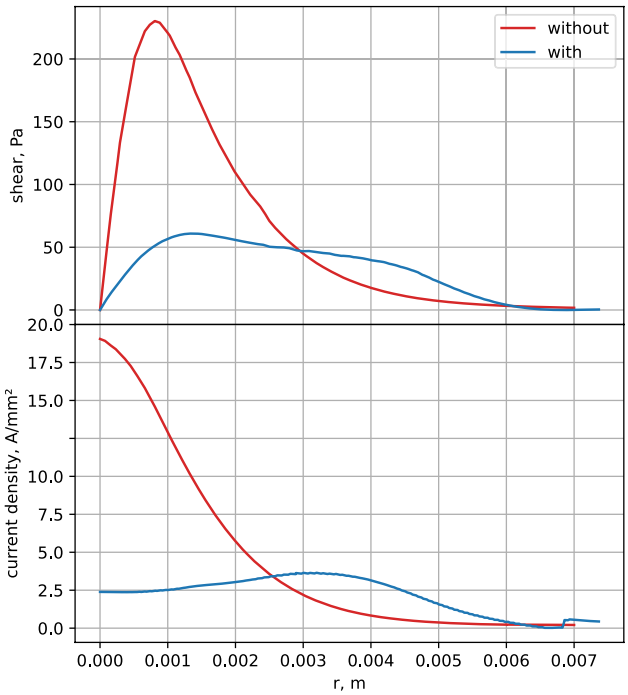


Fig. 15 Shear and current density without and with surface deformation at the anode-plasma interface at 250 A

This calibration significantly improves the prediction of the melt volume, as shown on the left side of Fig. 18. The new temperature measurements are displayed on the right side of this figure. Although the melt pool volume matches and the thermophysical properties of solid steel are known, the numerical temperatures are much higher, and the temporal evolution is different. In addition, the relative error in voltage increases from −5 to +12%, which implies to question this calibration methodology mainly used in literature.

7 Model recalibration discussion

Firstly, the previous results indicate no improvement in predicting the temporal evolution of temperatures. The heat input appears to be overestimated at the beginning of the process and underestimated at the end. Therefore, it turns out that using a constant ΔV_a is not right. Another mean would be to numerically introduce a temporal variation of ΔV_a, but with no physical justification. A more physical explanation could be to consider that the accumulation of metal vapors over time contributes to increase the heat input of the arc. Indeed, Zhao [9] shown that the vapors increase the voltage of the arc by changing its morphology mainly through an increase in radiation. This makes it a perspective for physically improving predictive accuracy of this model.

Secondly, an inconsistency between temperature measurements and melt volume is observed, which means that the interpretation of the micrographs can be questioned. It is assumed that the volume of the melt zone observed in the micrograph is the same as the volume of the melt pool at 10 s. However, this assumption implies perfect axisymmetry in the experimental fluid flows. Qualitative observations with a high-speed camera reveal significant agitation of particles floating on the surface of the melt pool. Their movements indicate substantial instability in the fluid flows, which could lead to small internal shaking of the melt pool,

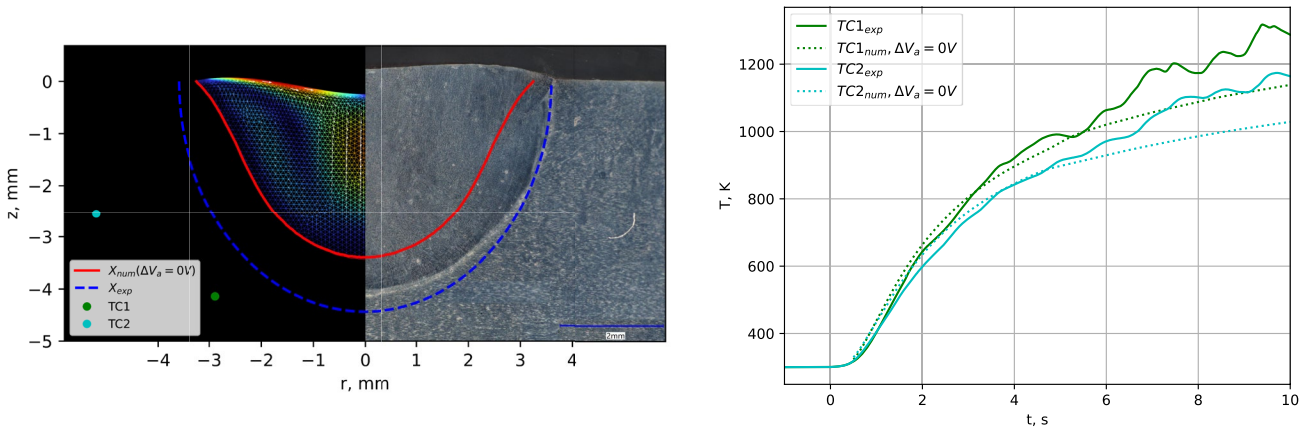


Fig. 16 Comparison of melt pools and thermograms before recalibration

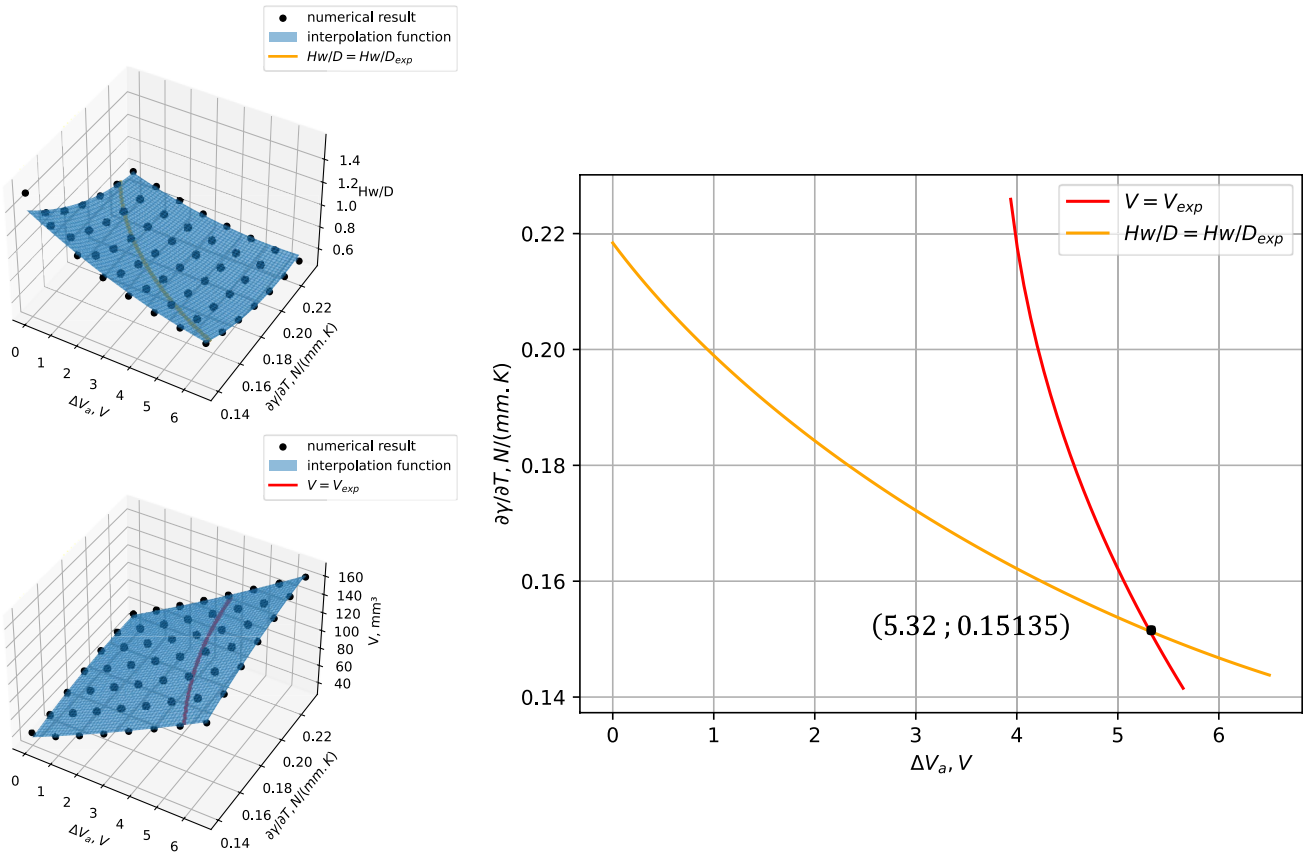


Fig. 17 Results of the calibration process optimisation of the couple $\left(\Delta V_a, \frac{\partial \gamma}{\partial T}\right)$ at 150 A

and therefore an increase in the experimental over all time fusion boundary. Taking this phenomenon into account in this model would involve using a 3D configuration, reducing the viscosity of liquid metal and adopting equations adapted to a turbulent regime.

8 Conclusion

The effect of increasing the intensity from 150 to 205 A is studied with both numerical and experimental approaches. The relative difference in the evolution of the width-depth

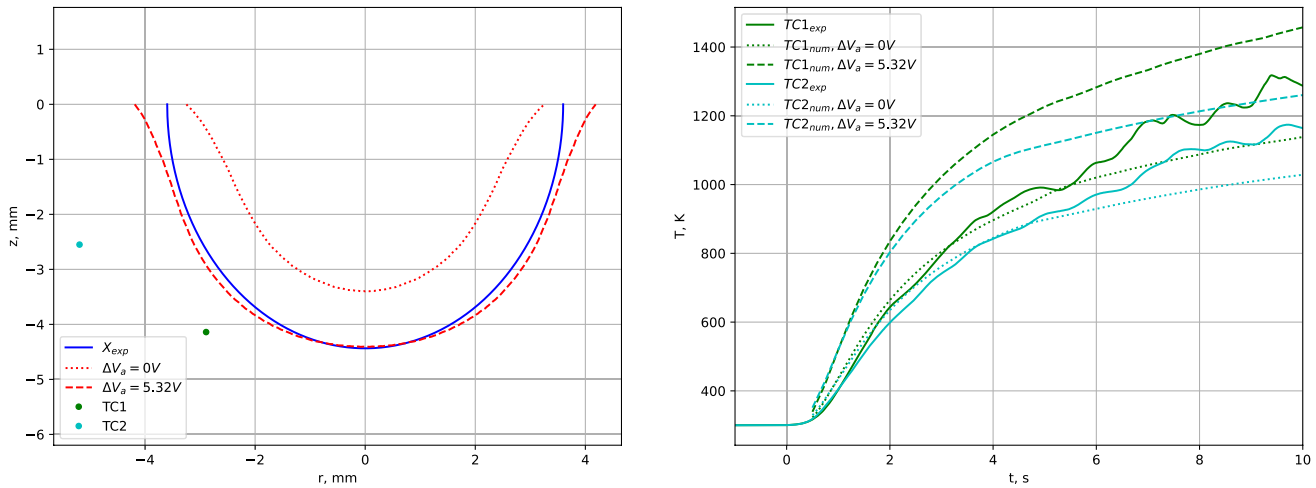


Fig. 18 Comparisons of melt pool shapes and thermograms after refit

ratio between simulation and experimental is equal to 7%, which is sufficient to explain the phenomena behind this trend and improve understanding of the process. Increasing the intensity amplifies the deformation of the anode-plasma interface by two. This has a direct impact on the arc by modifying the behavior of the plasma, resulting in a reduction in shear and a change in the current density at this interface. These two terms and the deformation of the melt pool surface have a strong influence through a radial ejection of the liquid metal, which explains the trends observed.

The numerical comparison at 150 A implies a numerical underestimation of 40% for the volume of the melt pool after $_{10\,\mathrm{s}}$ of welding. These differences justify the realization of a calibration to increase the heat input. A calibration methodology is used to have the same volume and shape of the experimental melt pool by adjusting the thermodependency coefficient ($\frac{\partial \gamma}{\partial T}$) and the anodic potential drop ($^{\Delta V_a}$). After this step, the difference in volume and shape is reduced to 2% but the temperatures within the anode are now overestimated by 10%. This result shows that the melt volume observed on the post-mortem micrography does not correspond exactly to the melt pool at the end of welding. This can be due to an internal oscillation of the experimental melt pool leading to a slight non-axisymmetric configuration unlike the simulation. This opens the possibility of increasing the complexity of fluid flow modeling within the melt pool. In addition, the evolution of heat input does not appear to be well described in simulation. The heat input increases after 5 s, which is not the case in simulation potentially due to the non-consideration of metal vapors.

Acknowledgements The authors acknowledge Ali Kansoun and François Portanguen for their help during the experimental phase.

Author contribution William Berckmans: investigation. Alexandre Brosse: writing—review and editing, supervision, and funding acquisition. Stephen Cadiou: writing—review and editing and supervision. Muriel Carin: writing—review and editing and supervision. Mickael Courtois: writing—review and editing and supervision. Marc Le Gal La Salle: writing—review and editing, writing—original draft, validation, software, methodology, investigation, formal analysis, data curation, and conceptualization.

Funding Open access funding provided by Université de Bretagne Sud. This work was supported by the ANRT (Association Nationale de la Recherche et de la Technologie) and Framatome.

Data availability All data that support the findings of this study are included within the article.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Lawson W, Kerr H (1976) Fluid motion in GTA weld pools. Part II: weld pool shapes. Weldg Res. Internation
2. Oreper GM, Eagar TW, Szekely J (1983) Convection in arc weld pools, Welding Res Supplement.
3. Szekely J (1989) Transport phenomena in welds with emphasis on free surface phenomena.
4. Zacharia T, David S, Vitek J, Debroy T (1989) Weld pool development during GTA and laser beam welding of type 304 stainless steel, part II—experimental correlation. Weld J 68:510s–519s
5. DebRoy T, David S (1995) Physical processes in fusion welding. Rev Mod Phys 67:85
6. Hsu KC, Etemadi K, Pfender E (1983) Study of the free-burning high-intensity argon arc. J Appl Phys 54:1293–1301. https://doi.org/10.1063/1.332195
7. McKelliget J, Szekely J (1986) Heat transfer and fluid flow in the welding arc. Metall Trans A 17:1139–1148
8. Tanaka M, Terasaki H, Ushio M, Lowke JJ (2002) A unified numerical modeling of stationary tungsten-inert-gas welding process. Metall Mater Trans A 33:2043–2052. https://doi.org/10.1007/s11661-002-0036-2
9. Zhao GY, Dassanayake M, Etemadi K (1990) Numerical simulation of a free-burning argon arc with copper evaporation from the anode, Plasma Chem Plasma Processing 10.
10. Lago F, Gonzalez JJ, Freton P, Gleizes A (2004) A numerical modelling of an electric arc and its interaction with the anode: part I. The two-dimensional model. J Phys D Appl Phys 37:883. https://doi.org/10.1088/0022-3727/37/6/013
11. Lin M, Eagar T (1983) Others, influence of surface depression and convection on arc weld pool geometry. Transport Phenomena Mater Processing 10:63–69
12. Traidia A, Roger F (2011) Numerical and experimental study of arc and weld pool behaviour for pulsed current GTA welding. Int J Heat Mass Transf 54:2163–2179. https://doi.org/10.1016/j.ijheatmasstransfer.2010.12.005
13. Nahed C, Gounand S, Medale M (2022) A numerical study of the effects of cathode geometry on tungsten inert gas type electric arcs. Int J Heat Mass Transf 182:121923. https://doi.org/10.1016/j.ijheatmasstransfer.2021.121923
14. Nestor OH (1962) Heat intensity and current density distributions at the anode of high current, inert gas arcs. J Appl Phys 33:1638–1648. https://doi.org/10.1063/1.1728803
15. Dupont JN, Marder A (1995) Thermal efficiency of arc welding processes, Weld J-Including Weld Res Suppl.
16. Brochard M (2009) Modèle couplé cathode-plasma-pièce en vue de la simulation du procédé de soudage à l'arc TIG, These de doctorat, Aix-Marseille 1. https://theses.fr/2009AIX11006. Accessed 9 Dec 2024
17. Kaddani A, Delalondre C, Simonin O, Minoo H (1994) Thermal and electrical coupling of arc electrodes. EDF
18. Kim CS (1975) Thermophysical properties of stainless steels, Argonne National Lab, Ill. (USA). https://doi.org/10.2172/4152287
19. Cadiou S, Courtois M, Carin M, Le Masson P, Guilmois L, Paillard P (2018) Development of a two-dimensional axial symmetry model for wire arc additive manufacturing, In: Mathematical Modelling of Weld Phenomena 12. Verlag Der Technischen Universität Graz. https://doi.org/10.3217/978-3-85125-615-4-56
20. Lowke JJ, Tanaka M (2006) "LTE-diffusion approximation" for arc calculations. https://doi.org/10.1088/0022-3727/39/16/017
21. Nguyen MC (2015) Modélisationet simulation multiphysique du bain de fusion en soudage à l'arc TIG, Université d'Aix-Marseille. https://theses.fr/2015AIXM4749
22. Bobkov V, Fokin L, Petrov E, Popov V, Rumiantsev V, Savvatimsky A (2008) Thermophysical properties of materials for nuclear engineering: a tutorial and collection of data. IAEA, Vienna, pp 18–21
23. Boulos MI, Fauchais P, Pfender E (2013) Thermal plasmas: fundamentals and applications, Springer Science & Business Media
24. Wu CS, Chen J, Zhang YM (2007) Numerical analysis of both front- and back-side deformation of fully-penetrated GTAW weld pool surfaces. Comput Mater Sci 39:635–642. https://doi.org/10.1016/j.commatsci.2006.08.018
25. Klapczynski V, Le Maux D, Courtois M, Bertrand E, Paillard P (2022) Surface tension measurements of liquid pure iron and 304L stainless steel under different gas mixtures. J Mol Liq 350:118558. https://doi.org/10.1016/j.molliq.2022.118558
26. Borel D, Delalondre C, Carpreau J, Cheron B, Boubert J (2014) Contribution to arc plasma modeling for welding TIG application, SNA + MC 2013 - Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo. https://doi.org/10.1051/snamc/201405117

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.